

Algorithms Problem Solving

Level 1

Hire Driver Case 2

The background of the slide is a grayscale photograph of a large, modern interior space, possibly a lobby or a large hall. The floor is made of large, light-colored square tiles arranged in a grid pattern. In the foreground and middle ground, there are silhouettes of several people. Two people are standing in the center, facing each other. In the foreground, there are three more people, some of whom appear to be sitting or leaning over a table, possibly engaged in a discussion or a meeting. The overall atmosphere is professional and collaborative.

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Definition

- ◆ This program reads a person's age, driving license status, and whether they have a recommendation, then determines whether they should be hired.
- ◆ The program uses a structure, functions, Boolean values, and conditional statements.
- ◆ A person can be hired either by meeting the normal hiring conditions or by having a recommendation.

Explanation

- ◆ The struct `stInfo` groups the user's related information:
 - age stores the person's age.
 - `hasDrivingLicense` stores whether the person has a driving license.
 - `hasRecommendation` stores whether the person has a recommendation.
- ◆ `readInfo()` reads the user's information and returns an `stInfo` object.
- ◆ `isValidInput()` checks the hiring conditions.
- ◆ If `hasRecommendation` is true, the function immediately returns true.
- ◆ If there is no recommendation, the function checks whether the person is older than 21 and has a driving license.
- ◆ `printResult()` calls `isValidInput()` and prints `Hired` if the result is true; otherwise, it prints `Rejected`.
- ◆ In `main()`, the functions are combined:
`printResult(readInfo());`

Steps in main()

1. Create the stInfo structure.
2. Read the user's age.
3. Read whether the user has a driving license.
4. Read whether the user has a recommendation.
5. Return the collected information.
6. Check if the user has a recommendation.
7. If there is a recommendation, return true.
8. Otherwise, check if age > 21 and hasDrivingLicense is true.
9. Print Hired if the result is true; otherwise print Rejected.

💡 Example

◆ If the user enters:

Age: 20

Driving license: 0

Recommendation: 1

The recommendation condition is true.

➡ Result

Hired

◆ If the user enters:

Age: 25

Driving license: 1

Recommendation: 0

The age and driving license conditions are true.

➡ Result

Hired

◆ If the user enters:

Age: 25

Driving license: 0

Recommendation: 0

The recommendation is false and the driving license condition is false.

➡ Result

Rejected

Note

- ◆ hasRecommendation is checked first. If it is true, the function returns true without checking the other conditions.
- ◆ The code uses `age > 21`, not `age >= 21`, so a person who is exactly 21 does not satisfy the normal hiring condition.

⊘ Warning

◆ Pay attention to the difference between $>$ and $\geq$. In this program, age must be greater than 21 for the normal hiring condition.



Comparison

Hired

- ✓ Has a recommendation.
- ✓ OR age is greater than 21 and has a driving license.
- ✓ Only one of these paths needs to be true.

Rejected

- ✓ No recommendation.
- ✓ AND the normal hiring conditions are not satisfied.
- ✓ The person is not hired.

✦✦ Summary

- ✓ A recommendation can allow the person to be hired without meeting the normal conditions.
- ✓ Without a recommendation, the person must be older than 21 and have a driving license.
- ✓ The program uses a structure and separate functions to organize the input, validation, and result.

Thank you 😊